

Data Discovery Report — APOE4 vs APOE3 organoid gene regulatory networks

⚠ Synthetic example. This report was generated by running the real pipeline against the mock collection built by `scripts/make_mock_inventory.py`. Every figure below is a genuine output of that run, but the underlying files, request IDs and paths are fabricated. It exists to show the format; it describes no real dataset.

Requested by: example user · **Date:** 2026-09-08 · **Run:** run-mock-example

Goal as understood:

Find existing data that could support a comparison of gene regulatory networks between APOE4 and APOE3 cerebral organoids, preferring processed matrices over raw reads.

Scope searched: collection MOCK Example Lab Collection · roots /lab_data/ · types all

Corpus: 154 files indexed · 133 matched by CSV · 12 inferred · 9 unattributed

1. Recommended datasets

Ordered by fit. Grouped by sequencing request, not by file.

Unattributed — multiome reanalysis directory

Fit: The only material in the collection that names both genotypes and a GRN output. Directly on target, but carries **no request record** — see the caveat below.

Project: — · **Requester:** unknown · **Status:** not in the CSV

Platform: not recorded

Original goal on file: none

PATH	ROLE	SIZE
/lab_data/reanalysis/apoe4_vs_apoe3_organoid_multiome/grn_edges.tsv	inferred GRN edge table	334 MB
/lab_data/reanalysis/apoe4_vs_apoe3_organoid_multiome/atac_peaks.bed	inferred ATAC peaks	544 MB

⚠ This directory reached the `inferred` tier at confidence 0.60 — its meaning comes from folder naming, not from a request record. Nothing verifies that these edges were computed from APOE4/APOE3 organoids, or which reference build was used. Confirm with whoever generated it before building on it.

MOLNG-1000 — Poly-A Stranded RNA-Seq

Fit: Bulk expression from the same lab, usable as a low-resolution sanity check on any GRN-derived expression claim. Not single-cell, so it cannot support network inference on its own.

Project: — · **Requester:** Garcia, Max · **Status:** Completed

Platform: HiSeq, Paired Reads

Original goal on file: not recorded

PATH	ROLE	SIZE
/lab_data/sequencing/abc/MOLNG-1000/H51EDRX3/counts/gene_counts.tsv	count matrix	500 MB
/lab_data/sequencing/abc/MOLNG-1000/H51EDRX3/align/sample.bam	alignments	500 MB

Shown 2 of 5 files. Full listing in `knowledge/ASSOCIATIONS.md`.

MOLNG-1049, MOLNG-1014, MOLNG-1119 — bulk RNA-Seq

Fit: Weaker matches on expression terms alone; none names a genotype. Listed for completeness, not recommended as a starting point.

Platform: HiSeq / Unknown, Paired Reads

PATH	ROLE	SIZE
/lab_data/sequencing/def/MOLNG-1049/H98STNMX2/counts/gene_counts.tsv	count matrix	100 MB
/lab_data/sequencing/ghi/MOLNG-1014/H57DTNMX2/counts/gene_counts.tsv	count matrix	400 MB
/lab_data/sequencing/jkl/MOLNG-1119/H81ADMX2/counts/gene_counts.tsv	count matrix	600 MB

2. Possibly relevant — inferred metadata only

These matched on folder naming, not on a request record. **Descriptions are inferred and unverified — confirm before relying on them.**

PATH	INFERRED AS	CONFIDENCE
/lab_data/reanalysis/apoe4_vs_apoe3_organoid_multiome/	APOE4/APOE3, cerebral organoid, 10x Multiome	0.60
/lab_data/reanalysis/sc_0E_h19/	single-cell, olfactory epithelium, H19	0.60
/lab_data/reanalysis/snRNA_VNO_adult_wt/	single-nucleus, vomeronasal organ, wild type	0.55

3. What these data cannot answer

3a. Unanswerable with any data in this collection

- **Differential GRN structure between genotypes at defined timepoints** — the one multiome directory records no timepoint, no replicate structure, and no genotype assignment per sample. *Requires:* the sample sheet for that reanalysis, or a re-run from the source matrices with genotypes declared.
- **Statistical power for edge-level comparison** — replicate count is unrecoverable from filenames alone. *Requires:* the original request record.

3b. Requested but never delivered

- **6 of 40 requests** have no result path recorded in the metadata CSV: MOLNG-1056, 1063, 1112, 1161, 1189, 1259. None names a genotype, so none is obviously relevant here — but absence of a path is not evidence of absence of data.

3c. Indexed but absent from the scanned scope

- **11 of 40 requests** have a CSV path but no matching files under `/lab_data/`: MOLNG-1007, 1028, 1084, 1126, 1140, 1168, 1196, 1231, 1245, 1266, 1273. These may be untransferred or outside the scanned roots. Worth checking before concluding the data does not exist.

3d. Present but undocumented

- **9 files (4.9 GB)** carry no attributable metadata — no request anchor, and folder names too vague to interpret. Largest clusters: `/lab_data/new_data/final2/` (3 files), `/lab_data/tmp/backup/v3/` (2 files), `/lab_data/scatch/` (2 files). These may contain relevant data; nothing can currently say so.

Standing limitations

- `Analysis Result Paths` is unpopulated for all 40 requests, so no derived analysis output can be matched by request ID. Processed results — including the multiome directory above — reach the `inferred` tier at best.
- The metadata CSV indexes sequencing requests only. Proteomics, imaging, EM, behavior and ephys have no request number and cannot appear in it. Absence from the CSV is not absence from the collection.
- All requests are from one lab; no cross-lab comparison is supported.

4. Suggested next steps

- Identify the author of `/lab_data/reanalysis/apoe4_vs_apoe3_organoid_multiome/` and recover its sample sheet — this single artifact decides whether the question is answerable today.
 - Check whether the 11 requests in §3c were transferred, before commissioning anything new.
 - Add a README to `/lab_data/new_data/` — 9 files are currently unattributable.
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Generated by the Globus discovery agent. Associations recorded in `knowledge/file_associations.jsonl` ; this run logged in `knowledge/query_log.jsonl` . Corrections welcome — they become permanent overrides.